

# ANIL PRADHAN

BACKEND ENGINEER | LLM APPLICATION DEVELOPER

Backend-focused Computer Science undergraduate with hands-on experience building production-grade AI applications using FastAPI, LangGraph, RAG, Docker, and cloud technologies. Passionate about scalable backend architecture, distributed systems, and AI-powered software.

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Bhubaneswar, Odisha

LinkedIn | GitHub | Portfolio



## EDUCATION

### Institute of Technical Education & Research (ITER)

Siksha 'O' Anusandhan (SOA) University, Bhubaneswar

Bachelor of Technology (B.Tech), Computer Science & Engineering

2024 - 2028 | CGPA: 9.11/10

### Council of Higher Secondary Education (CHSE), Odisha

Class XII (PCM with IT)

2022 - 2024 | 85%



## EXPERIENCE

### College Ambassador (Corporate Projects Program) | Techfest, IIT Bombay

Jul 2026 - Present

- Selected as a College Ambassador for Techfest, IIT Bombay, representing one of Asia's largest science and technology festivals.
- Completed industry-sponsored corporate projects across web development, business strategy, and market research in collaboration with partner organizations.
- Qualified for Round 2 of the FlowFinance Growth Strategy & Business Development Challenge by developing a business strategy presentation and market analysis.
- Collaborated with multidisciplinary teams while delivering project submissions within defined deadlines.



## TECHNICAL PROJECTS

### AI Career Mentor | Live Demo | GitHub

Mar 2026 - Present

Solo Full Stack Developer

- Architected a parallel LangGraph workflow with 7 specialized AI agents, reducing end-to-end career analysis latency by 60% (4 min → 60 sec).
- Built a real-time AI Mock Interview Engine using FastAPI, WebSockets, and a 7-stage Finite State Machine (FSM), delivering company-specific technical interviews with live coding evaluation and personalized feedback.
- Developed "Anyra", a real-time multimodal AI Voice Coach powered by the Gemini Live API, enabling full-duplex voice conversations with resume-aware career guidance, interview coaching, and personalized learning support.
- Engineered a fault-tolerant multi-LLM architecture (Groq, NVIDIA NIM, Cerebras) with circuit breakers, automatic failover, and rate limiting, achieving uninterrupted service availability.
- Optimized RAG retrieval using local ONNX embeddings with keyword-search fallback, eliminating embedding API dependency while ensuring reliable performance on low-memory cloud deployments validated through 116 automated tests.

Tech: FastAPI • LangGraph • Next.js • ChromaDB • Redis • Docker • Gemini Live • ONNX

### TaskPilot AI | Live Demo | GitHub

Jun 2026 - Jul 2026

Integration Lead | Team IdeaForge-E

- Architected a multi-agent engineering productivity platform integrating 7 fragmented developer tools (Jira, GitHub, Slack, Email, Calendar, Meetings, and Incidents), reducing manual task tracking by ~2 hours/day.
- Designed a hybrid AI task extraction pipeline combining heuristic rules, fuzzy matching, and LLM reasoning (Qwen), recovering 35% more hidden action items while reducing LLM inference costs by 75% through selective AI execution.
- Built a concurrency-safe orchestration backend using FastAPI, SQLite WAL mode, and Redis, enabling reliable parallel multi-agent execution with real-time P1 incident reprioritization in under 15 seconds.
- Developed a self-healing workflow engine with automated stale-run recovery and an AI-powered quality evaluator that validated task completeness, actionability, and generated context-aware follow-up questions.

Tech: FastAPI • LangGraph • React • SQLite • Redis • LLMs



## TECHNICAL SKILLS

Languages: Python, Java, JavaScript, SQL

Backend: FastAPI, LangGraph, Pydantic, SQLAlchemy 2.0, Websocket, Prometheus, Sentry, pdfplumber, edge-tts

Databases: Neon Postgres, SQLite, Upstash Redis, ChromaDB (Vector Store)

DevOps & Infra: Docker, GitHub Actions (CI/CD), Vercel, Render.com

Frontend: HTML, CSS, Tailwind CSS, Axios, React

AI / LLM & RAG: Groq Cloud, Cerebras Cloud, NVIDIA NIM, Gemini Live API, Tavily, Serper.dev, DuckDuckGo

Auth & Security: JWT (python-jose), Google OAuth 2.0, bcrypt, CORS Middleware, Rate Limiting, HTTPS

Tools: VS Code, Git, GitHub, Swagger UI, Postman



## ACHIEVEMENTS & CERTIFICATIONS

- Google Cloud Arcade Program (2024-25): Completed GCP challenges, earning 75 Arcade Points across hands-on cloud labs.
- Solved 200+ problems across GFG and LeetCode, awarded GFG Certificate and T-shirt.
- Certifications: Credentials in Python, AI, and Web Dev from IBM, Cisco, Anthropic, Infosys, Udemy.
- Competitions: Participated in multiple IIT/NIT coding contests and hackathons focused on AI systems, backend engineering, and algorithmic problem solving.
- Hackathons: Built production-grade MVP solutions for Microsoft AI DevDays (\$80K+), Amazon Nova AI (\$95K+), and Google Solution Challenge 2026.
- Active participant in Developer Communities (GDG Events) and college tech events, exploring AI systems, cloud infrastructure, and backend engineering.